

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2026.0000000

Contact-Model-Aware Sensorless Force Estimation and Compliant Predictive Control for Velocity-Actuated Low-Cost Manipulators

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The research was funded by Universidad Tecnológica Indoamérica.

ABSTRACT Compliant physical interaction on low-cost manipulators must work without a force/torque sensor, whose price exceeds that of the arm, and with servos that accept velocity rather than torque commands. Residual-based sensorless estimation was developed for torque-actuated arms with six or more joints, where neither constraint applies, and this paper characterizes what changes when both do. We first show that for fewer than six joints the contact model, not the Jacobian rank, determines what can be sensed. Inverting the residual for a full wrench is under-determined and silently reassigns contact force to a spurious moment, discarding 50.5 % of an applied unit force on average over 2197 configurations of a 4-DoF arm and up to 99.6 %, while still correlating with ground truth at 0.997. A point-contact inversion is exact wherever the linear-velocity Jacobian has full row rank, which holds throughout the sampled workspace, and reduces force RMSE by a factor of fourteen against an independent physics engine. We then show that velocity-level actuation, usually regarded as an obstacle, restricts the dynamic model to the estimator alone: the residual is invariant to the inner servo loop, the finite-differenced acceleration carries 0.006 % of the signal, and a kinematic controller using no inertia matrix, bias term, or servo gain matches a torque-level formulation throughout the compliant loop at one twenty-fifth of the computational cost. Under the quantization and stiction of the real servo, over 2000 randomized trials, the point-contact inversion attains 18.0 % median relative error against 69.6 %, and rejecting postures with $\sigma_{\min}(\mathbf{J}_v) < 0.06$ halves its ninetieth percentile while leaving the six-dimensional error unchanged, separating amplified noise from structural misattribution. Finally, because the actuator can exert only 3.9 to 5.9 N against its worst direction while manual guidance requires 5 to 15 N, compliance is synthetic below that threshold and physical above it, with a measurable transition. All results are obtained in simulation against a model validated to 10^{-10} ; a staged hardware protocol with quantified detection power is specified.

INDEX TERMS Compliant control, force estimation, low-cost robotics, model predictive control, physical human-robot interaction, sensorless sensing, under-actuated manipulators, velocity control.

I. INTRODUCTION

COMPLIANT physical interaction, in which a manipulator yields to an external push and recovers its task once released, underpins hand guiding, contact-rich assembly, and safe operation in shared workspaces. The canonical implementation measures the interaction wrench with a wrist force/torque (F/T) sensor and closes an impedance or admit-

tance law around it [1], [2].

On research-grade manipulators this is unremarkable. On arms assembled from commercial smart servos, a class that increasingly serves education, small-scale automation, and mobile manipulation research, it is not viable. A six-axis F/T sensor typically costs more than the entire mechanism, and mounting one consumes payload and structural stiffness that

a lightweight arm cannot spare. The economic argument that motivates a low-cost arm is precisely the argument against instrumenting it.

Two obstacles are usually cited against doing without the sensor on such a platform, and this paper argues that both have been misread.

A. FIRST OBSTACLE: TOO FEW JOINTS TO SENSE A WRENCH

The established sensorless alternative is a momentum or residual observer, which compares measured joint accelerations and actuator torques against a dynamic model and attributes the mismatch to an external wrench [3]–[5]. The approach is mature and requires no hardware beyond what a servo already reports; [6] surveys its detection, isolation, and identification variants.

The residual lives in joint space and is mapped to a Cartesian wrench through the transpose Jacobian pseudoinverse. This step recovers six unknowns from n measurements. For $n \geq 6$ it is at worst a conditioning question. For the four-joint arms that dominate the low-cost segment the system is under-determined by construction, and the minimum-norm solution resolves the ambiguity by distributing the residual between force and moment. When the physical contact is a pure force, as a human push is, part of it is returned as a spurious moment and is absent from the force estimate.

The failure is silent and easy to misdiagnose. Applying a 6 N push to the arm studied here and inverting in six dimensions returns 0.99 N, an error of 83 %, which reads as evidence that a four-joint arm cannot perceive that interaction. The joint torques say otherwise: the same push produces $1.29 \text{ N} \cdot \text{m}$ at the base joint. The information is present in the measurement and the estimator discards it.

B. SECOND OBSTACLE: THE ACTUATOR TAKES VELOCITY, NOT TORQUE

Smart servos of this class are commanded in velocity, and the residual observer is formulated in torque. The apparent mismatch is often taken to mean that sensorless estimation is unavailable on such hardware, or that the servo loop must first be identified.

Neither is the case, and the reason is that the servo exposes a third signal besides position and velocity: the effort it is applying, reported as motor current. That channel is exactly the quantity the residual requires, and it takes the place of the force/torque sensor throughout this work. Since the residual consumes the applied torque as measured rather than as modeled, the internal velocity loop may remain a black box.

Section V works out what this changes, and the consequences run in the favorable direction: the interaction signal migrates from the acceleration to the actuator torque, the dynamic model becomes unnecessary in the controller, and the actuator saturates below the force range of manual guidance, so the arm yields physically without any control effort.

TABLE 1. Residual-based interaction sensing: operating regime of representative approaches.

	joints	actuation	inversion	force error
[3]	≥ 6	torque	six-dimensional	—
[4]	7	torque	six-dimensional	—
[5]	7	torque	point contact	at one pose
[6]	≥ 6	torque	six-dimensional	—
this work	4	velocity	point contact	over workspace

C. CONTRIBUTIONS

As shown in Table 1, residual-based interaction sensing has been developed for torque-actuated manipulators with six or more joints, where the wrench inversion is over-determined and the actuator accepts the quantity the controller computes. Neither condition holds on a low-cost smart-servo arm. To close this gap, this paper makes the following three contributions:

- We characterize sensorless force estimation for $n < 6$ and show that the contact model, not the Jacobian rank, determines what can be sensed. Over 2197 workspace configurations the standard six-dimensional inversion discards 50.5 % of an applied unit force on average and up to 99.6 %, and it fails silently, correlating with ground truth at 0.997 while carrying a 2.227 N error. Sweeping the full sphere of push directions at six configurations, its error is a linear function of the alignment with the least observable direction with correlation 0.9993, whereas a point-contact inversion is exact to 4×10^{-13} % over all 12 000 directions and reduces force RMSE by a factor of fourteen against an independent physics engine.
- We show that under velocity-level actuation the dynamic model is required by the estimator alone, so the controller can be model-free. Of the three possible formulations, the dynamically consistent velocity-level problem is ill-posed unless the servo loop architecture is identified and fails at every solve, whereas a kinematic controller using neither inertia matrix, bias term, nor servo gain matches the torque-level formulation throughout the compliant loop at one twenty-fifth of the computational cost. What must be identified on hardware reduces to the mass properties and one torque constant, both serving estimation only.
- We evaluate both inversions under the quantization and stiction of the real servo, over 2000 randomized configurations and contact forces. The point-contact inversion attains 18.0 % median relative error against 69.6 % for the six-dimensional one, and rejecting postures with $\sigma_{\min}(\mathbf{J}_v) < 0.06$ halves its ninetieth percentile from 61 % to 30 % while leaving the six-dimensional error unchanged, confirming that one error is amplified noise and the other is structural misattribution.

Two of the ingredients are individually elementary and we do not claim them: that $\mathbf{J}^T \mathbf{F} = \boldsymbol{\tau}_{\text{ext}}$ is under-determined for $n < 6$ is linear algebra, and restricting the unknown to a force at a known contact point is standard practice [5]. What is new

is the regime, the quantification, and what follows from them for this class of hardware.

D. SCOPE AND ORGANIZATION

All results reported here are obtained in simulation against a model validated to 10^{-10} against an independent physics engine, which also serves as the plant for the interaction experiments. Hardware validation is not claimed; Section VIII states the limitations. Section II positions the work, Section III describes the platform and model, Sections IV to VI develop the method, Section VII reports experiments, and Section IX concludes.

II. RELATED WORK

A. SENSORLESS INTERACTION SENSING

Residual-based external wrench estimation originates in collision detection. De Luca and Mattone introduced the sensorless residual for hybrid force/motion control [3], and the method was extended to safe reaction on lightweight arms [4]. Magrini *et al.* developed the virtual force sensor, which for redundant arms estimates the contact location along the structure in addition to the force [5]. A comprehensive account of detection, isolation, and identification appears in [6].

This literature is developed almost exclusively for manipulators with six or seven joints, where the transpose Jacobian has full column rank and the wrench inversion is overdetermined. The under-actuated regime, in which the number of joints is smaller than the dimension of the wrench, receives comparatively little attention, and the consequences for the inversion step are, to our knowledge, not characterized. Nor is the interaction between sensorless estimation and velocity-level actuation, which is the norm on low-cost hardware. This paper addresses both gaps.

The closest prior work is [5], and the distinction is worth stating precisely because both use a point-contact model. There, the contact location is the *unknown* to be estimated, on a seven-joint arm where the six-dimensional inversion is already well posed and the point-contact assumption serves to localize the contact along the structure. Here the location is known by construction, the unknown is the force alone, and the point-contact assumption serves a different purpose: it reduces the dimension of the unknown from six to three, which is what makes the problem solvable at all when $n = 4$. The distinction matters because in the redundant, torque-actuated setting the six-dimensional inversion is merely one option among several, whereas for $n < 6$ it is structurally wrong in the sense quantified in Section IV.

B. CAPABILITY ANALYSIS OF MANIPULATORS

Manipulability ellipsoids [7] and force polytopes [8] describe the set of velocities or forces a manipulator can *produce*, and are standard tools for posture optimization. The question posed here is the dual one, namely which external contacts a manipulator can *perceive* through its own actuators. We show that for $n < 6$ the answer is governed not by the rank deficit of the full Jacobian, as a direct transposition

of the classical analysis would suggest, but by the contact model the estimator assumes. The force polytope reappears in Section V-C in its classical role, where it sets the threshold between synthetic and physical compliance.

C. PREDICTIVE CONTROL WITH A COUPLED POSE ERROR

Tracking a pose in SE(3) requires a scalar error that penalizes translation and rotation jointly. The simplest choice keeps them decoupled, adding a position error and an orientation error independently. A coupled alternative expresses the translational part in the frame of the rotational error through the inverse left Jacobian of SO(3) [9]. Section VII-I reports an ablation isolating what that coupling contributes on an under-actuated arm, where the reference is generically unreachable. Real-time nonlinear predictive control is enabled by the real-time iteration scheme [10] and by embedded solver frameworks [11]–[13].

III. SYSTEM DESCRIPTION AND MODELING

A. PLATFORM

The manipulator has $n = 4$ revolute joints driven by Dynamixel MX-28R smart servos, with a rigid gripper treated as a fixed payload. Actuation limits are $|\tau_i| \leq 1.4 \text{ N} \cdot \text{m}$ and $|\dot{q}_i| \leq 5.969 \text{ rad/s}$. Each servo reports joint position and velocity, and its current measurement provides a proxy for the applied actuator torque. The servo accepts a velocity command; there is no torque command and no F/T sensor.

The first joint is a base yaw with vertical axis, and the remaining three are pitch joints whose axes are mutually parallel and orthogonal to it. This layout, a yaw base followed by a planar arm, is the standard low-cost architecture and is what determines the structure of the estimation problem.

B. RIGID-BODY DYNAMICS

The joint-space dynamics under an external wrench $\mathbf{F}$ applied at the end-effector are

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{h}(\mathbf{q}, \dot{\mathbf{q}}) = \boldsymbol{\tau} + \mathbf{J}(\mathbf{q})^\top \mathbf{F}, \quad (1)$$

with $\mathbf{q} \in \mathbb{R}^4$ the joint angles, $\boldsymbol{\tau} \in \mathbb{R}^4$ the actuator torques, and

$$\mathbf{J}(\mathbf{q}) = \begin{bmatrix} \mathbf{J}_v(\mathbf{q}) \\ \mathbf{J}_\omega(\mathbf{q}) \end{bmatrix} \in \mathbb{R}^{6 \times 4} \quad (2)$$

the geometric end-effector Jacobian partitioned into linear and angular blocks. The wrench is correspondingly $\mathbf{F} = [\mathbf{f}^\top \mathbf{m}^\top]^\top$, with $\mathbf{f}$ the contact force and $\mathbf{m}$ the contact moment in the world frame. The inertia matrix is assembled from per-link geometric Jacobians,

$$\mathbf{M} = \sum_i \left(m_i \mathbf{J}_{v_i}^\top \mathbf{J}_{v_i} + \mathbf{J}_{\omega_i}^\top \mathbf{R}_i \mathbf{I}_i \mathbf{R}_i^\top \mathbf{J}_{\omega_i} \right) + \mathbf{D}_a, \quad (3)$$

where $\mathbf{D}_a$ is the servo rotor armature, and $\mathbf{h}$, collecting Coriolis, centrifugal, and gravitational terms, follows from the Lagrangian.

C. SENSING AND ACTUATION INTERFACE

The servo exposes three measurements and accepts one command, and the way they are used is the whole architecture, shown in Fig. 1.

- *Position* $\mathbf{q}$ and *velocity* $\dot{\mathbf{q}}$, from the internal encoder. The acceleration $\ddot{\mathbf{q}}$ is obtained by finite differences of $\dot{\mathbf{q}}$.
- *Effort*, reported as motor current and converted to the applied actuator torque $\boldsymbol{\tau}_{\text{act}} = \mathbf{K}_t \mathbf{i}$ through a per-joint constant $\mathbf{K}_t$. *This third signal is what replaces the force/torque sensor.* It is the only channel through which the interaction reaches the controller, and Section V shows that under velocity-level actuation it carries essentially the entire interaction signal.
- *Command* $\mathbf{u}$, a joint velocity setpoint. There is no torque command.

D. THE VELOCITY-ACTUATED PLANT

Since the actuator is commanded in velocity, the plant seen by the controller is not (1). Modeling the internal servo loop as a proportional velocity regulator,

$$\boldsymbol{\tau} = k_v (\mathbf{u} - \dot{\mathbf{q}}), \quad (4)$$

and substituting into (1) gives the closed-loop plant in compact form,

$$\mathbf{M}(\mathbf{q}) \ddot{\mathbf{q}} + k_v \dot{\mathbf{q}} + \mathbf{h}(\mathbf{q}, \dot{\mathbf{q}}) = k_v \mathbf{u} + \mathbf{J}(\mathbf{q})^\top \mathbf{F} \quad (5)$$

which we verified against (1) to 4.3×10^{-14} over randomized states, gains, and wrenches. Equation (5) makes the role of the servo explicit: it contributes a damping term $k_v \mathbf{I}$ and an input gain k_v , and as $k_v \rightarrow \infty$ it degenerates to the kinematic limit $\dot{\mathbf{q}} = \mathbf{u}$ in which the arm is a pure velocity source. Actuator saturation enters as $\|\boldsymbol{\tau}\|_\infty \leq \tau_{\text{max}}$ applied to (4), and Section V-C shows that this bound, rather than the loop gain, is what governs the interaction.

Remark 1 (The estimator does not use k_v). *Equation (4) is written for analysis only. The residual (7) uses $\boldsymbol{\tau}_{\text{act}}$ as measured through the effort channel, not as modeled, so k_v appears nowhere in the estimator. We verified that the recovered force is identical for $k_v \in \{1, 10, 100\}$. The internal servo loop may therefore remain a black box, and no system identification of it is required. This is the formal reason why velocity-level actuation does not obstruct sensorless estimation.*

E. POSE ERROR

Tracking requires a scalar measure of pose error. We use a six-dimensional error whose rotational part is the axis-angle vector $\boldsymbol{\varphi}$ of the orientation error and whose translational part is expressed in the frame of that rotational error,

$$\mathbf{e} = \begin{bmatrix} \boldsymbol{\varphi} \\ \boldsymbol{\rho} \end{bmatrix} \in \mathbb{R}^6, \quad \boldsymbol{\rho} = \mathbf{J}_\ell(\boldsymbol{\varphi})^{-1} \mathbf{t}_{\text{err}}, \quad (6)$$

with $\mathbf{J}_\ell$ the left Jacobian of SO(3) [9] and $\mathbf{t}_{\text{err}}$ the translational error. The factor $\mathbf{J}_\ell^{-1}$ is what distinguishes (6) from a decoupled error that adds position and orientation independently,

TABLE 2. Model validation against MuloCo, eight configurations.

quantity	maximum deviation
inertia matrix $\mathbf{M}$	8.8×10^{-11}
bias term $\mathbf{h}$	2.7×10^{-9}
forward kinematics, position	6.9×10^{-13} m
forward kinematics, orientation	0 (machine precision)
geometric Jacobian $\mathbf{J}$	4×10^{-17}

and since $\mathbf{J}_\ell^{-1} \rightarrow \mathbf{I}$ as $\|\boldsymbol{\varphi}\| \rightarrow 0$ the two agree in the small-error limit. Section VII-I quantifies what the coupling buys.

F. MODEL VALIDATION

The symbolic model was built independently of the simulator and validated against MuJoCo [14] over eight configurations spanning the joint ranges (Table 2). The agreement is at the numerical precision of the reference implementation, so modeling error is not a confounding factor in what follows. This matters, because the central claim of this paper is that an observed estimation error belongs to the inversion step, and that attribution is credible only if the model itself is exact.

IV. CONTACT-MODEL-AWARE FORCE ESTIMATION

A. THE RESIDUAL AND THE ESTIMATION PROBLEM

Rearranging (1) gives the measurable residual

$$\boldsymbol{\tau}_{\text{ext}} = \mathbf{M}(\mathbf{q}) \ddot{\mathbf{q}} + \mathbf{h}(\mathbf{q}, \dot{\mathbf{q}}) - \boldsymbol{\tau}_{\text{act}} = \mathbf{J}(\mathbf{q})^\top \mathbf{F} \in \mathbb{R}^4, \quad (7)$$

computed from joint measurements alone: $\mathbf{q}$ and $\dot{\mathbf{q}}$ from encoders, $\ddot{\mathbf{q}}$ by finite differences, and $\boldsymbol{\tau}_{\text{act}}$ from the servo current. Everything the arm knows about the interaction is in these four numbers.

Problem 1. *Given $\boldsymbol{\tau}_{\text{ext}} \in \mathbb{R}^4$ from (7), recover the external contact force acting at the end-effector, and characterize the configurations in which recovery is possible and the accuracy attainable.*

Problem 1 asks for a *force*, not a wrench. That distinction is the substance of this section: the number of unknowns admitted into the inversion determines whether the problem is solvable, and the conventional choice admits too many.

B. THE SIX-DIMENSIONAL INVERSION

The conventional estimator treats the full wrench as unknown and solves $\mathbf{J}^\top \mathbf{F} = \boldsymbol{\tau}_{\text{ext}}$ for $\mathbf{F} \in \mathbb{R}^6$ from four equations. The system is under-determined and the minimum-norm solution is

$$\hat{\mathbf{F}}_6 = \mathbf{J}(\mathbf{J}^\top \mathbf{J})^{-1} \boldsymbol{\tau}_{\text{ext}} = \mathbf{P} \mathbf{F}, \quad \mathbf{P} = \mathbf{J}(\mathbf{J}^\top \mathbf{J})^{-1} \mathbf{J}^\top, \quad (8)$$

with $\mathbf{P}$ the orthogonal projector onto $\text{col } \mathbf{J} \subset \mathbb{R}^6$.

Proposition 1 (Force-to-moment misattribution). *For a pure contact force $\mathbf{F} = [\mathbf{f}^\top \mathbf{0}^\top]^\top$, the force returned by (8) is $\mathbf{P}_{\text{ff}} \mathbf{f}$, where $\mathbf{P}_{\text{ff}}$ is the leading 3×3 block of $\mathbf{P}$. In general $\|\mathbf{P}_{\text{ff}} \mathbf{f}\| < \|\mathbf{f}\|$, and the deficit appears as a nonzero moment in the lower block of $\hat{\mathbf{F}}_6$.*

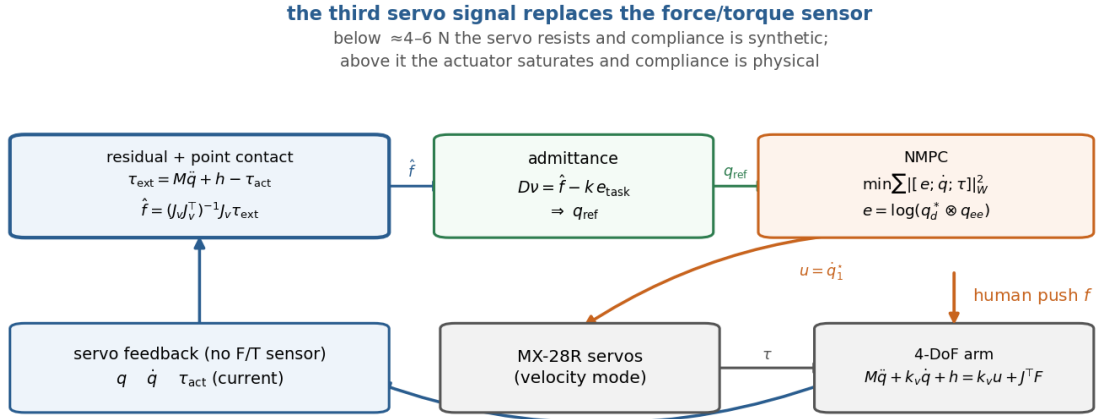


FIGURE 1. System architecture. The servo returns position, velocity, and effort; the effort channel takes the place of a force/torque sensor. The residual and the point-contact inversion produce $\hat{f}$, an admittance converts it into a displaced pose reference, and the predictive controller returns a joint velocity command, which is the only input the actuator accepts.

The mechanism is misattribution, not loss of information. A pure force and a suitable force-moment pair produce identical joint torques, and minimum-norm selects the latter because it has smaller Euclidean norm in $\mathbb{R}^6$. Both are consistent with the measurement; only one is realizable by the contact under consideration, and nothing in the formulation encodes which.

At the task configuration $\mathbf{q}_{\text{task}} = [0, 1.2, -0.9, 0.4]$ rad, a 6 N push normal to the plane of the arm produces $\boldsymbol{\tau}_{\text{ext}} = [1.2867, 0, 0, 0]^T$ N·m. Equation (8) returns 0.264 N of force together with 1.230 N·m of moment. The eigenvalues of $\mathbf{P}_{ff}$ there are $\{0.044, 1, 1\}$: two force directions pass unaltered while a third is almost entirely converted into a moment.

Remark 2 (Aggregate observability is uninformative). *Since $\mathbf{P}$ is an orthogonal projector, $\text{tr } \mathbf{P} = \text{rank } \mathbf{J}$ identically, so for wrench directions drawn uniformly on the unit sphere of $\mathbb{R}^6$ the expected fraction returned by (8) is $\text{rank } \mathbf{J} / 6$ regardless of configuration. Any workspace-averaged observability index built on $\mathbf{P}$ is constant by construction and carries no information about posture. The structure lies in how the deficit is distributed across directions.*

Remark 3 (Numerical care at singularities). *At singular configurations $\mathbf{J}$ loses rank and (8) is undefined. Throughout this work $\mathbf{P}$ is computed from the thin singular value decomposition $\mathbf{J} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T$ as $\mathbf{P} = \mathbf{U}_r\mathbf{U}_r^T$ with r the numerical rank, which remains valid there.*

C. THE POINT-CONTACT INVERSION

A human push at the end-effector transmits a force at a known point and no moment about that point. Imposing $\mathbf{m} = \mathbf{0}$ in (7) leaves

$$\mathbf{J}_v(\mathbf{q})^T \mathbf{f} = \boldsymbol{\tau}_{\text{ext}}, \quad \mathbf{J}_v^T \in \mathbb{R}^{4 \times 3}, \quad (9)$$

over-determined: four equations in three unknowns, with least-squares solution

$$\hat{\mathbf{f}}_3 = (\mathbf{J}_v \mathbf{J}_v^T)^{-1} \mathbf{J}_v \boldsymbol{\tau}_{\text{ext}}. \quad (10)$$

Proposition 2 (Exact recovery). *$\hat{\mathbf{f}}_3 = \mathbf{f}$ for every pure contact force applied at the end-effector if and only if $\text{rank } \mathbf{J}_v = 3$. The result is independent of the rank deficit of $\mathbf{J}$ and does not involve $\mathbf{J}_\omega$.*

Proof. By construction (9) is consistent, since $\boldsymbol{\tau}_{\text{ext}} = \mathbf{J}_v^T \mathbf{f}$ for the true $\mathbf{f}$. A consistent least-squares problem attains zero residual, and its solution is unique precisely when $\mathbf{J}_v^T$ has trivial kernel, that is when $\text{rank } \mathbf{J}_v = 3$. If $\text{rank } \mathbf{J}_v < 3$ the solution set is an affine subspace of dimension $3 - \text{rank } \mathbf{J}_v$, and (10) returns its minimum-norm element, which differs from $\mathbf{f}$ whenever $\mathbf{f}$ has a component in $\ker \mathbf{J}_v^T$. The angular block $\mathbf{J}_\omega$ appears nowhere in (9), so the rank of $\mathbf{J}$ is irrelevant. $\square$

Proposition 2 moves the governing condition from $\mathbf{J} \in \mathbb{R}^{6 \times 4}$, which cannot attain rank 6, to $\mathbf{J}_v \in \mathbb{R}^{3 \times 4}$, which generically attains rank 3. Four joints are not too few to sense a contact force at the end-effector; they are one more than the minimum required.

Remark 4 (What the model assumption costs). *The point-contact model requires the contact location to be known. This holds for hand guiding at the end-effector and fails for contact along the links. If the contact additionally transmits a moment, for instance through a rigidly grasped object, the ambiguity of Proposition 1 returns and (8) becomes the honest choice, with the accompanying loss. Extending the estimator to an unknown contact point, in the spirit of [5], is future work.*

D. CONDITIONING AS THE BINDING LIMITATION

With exactness settled, the practically relevant quantity is sensitivity to measurement noise. Perturbing the residual by $\delta\tau$ perturbs the estimate by

$$\|\delta\hat{f}_3\| \leq \frac{\|\delta\tau\|}{\sigma_{\min}(\mathbf{J}_v)}, \quad (11)$$

so $\sigma_{\min}(\mathbf{J}_v)$ bounds the amplification of residual error into force error and replaces observability as the quantity to map.

E. WORKSPACE CHARACTERIZATION

We sweep the three pitch joints over their full ranges on a 13^3 grid, fixing the base yaw since it only rotates the picture in the world frame and leaves every quantity of interest invariant. This gives 2197 configurations, at each of which we compute the rank and singular values of $\mathbf{J}_v$ and apply a random unit contact force, recording the error of both estimators. Table 3 summarizes and Fig. 2 maps the outcome.

Two facts stand out. First, $\text{rank } \mathbf{J}_v = 3$ in 100 % of the grid, so by Proposition 2 the contact force is exactly recoverable everywhere, and the numerical error of $\hat{f}_3$ is at machine precision. Second, over the same set the six-dimensional inversion discards 50.5 % of the applied force on average and, in the worst configuration and direction, essentially all of it.

The geometric reason the six-dimensional inversion fails in a particular direction is worth stating, because it explains why the effect is systematic on this class of arm. The three pitch joints are coplanar, so the torques they produce resist only loads in the plane of the arm. A force component normal to that plane reaches the actuators through a single path, the moment it exerts about the base yaw axis. A moment applied directly about that same axis produces the identical joint torque with a smaller norm in $\mathbb{R}^6$, so the minimum-norm criterion prefers it. The out-of-plane force direction is therefore exactly the one that the six-dimensional inversion converts into a moment, in every configuration of this architecture.

F. DIRECTIONAL CHARACTERIZATION

The preceding argument predicts that the error of the six-dimensional inversion is governed by the *direction* of the push and by nothing else. That prediction is falsifiable and we test it directly, since a claim of this kind cannot rest on the two push directions used in the closed-loop experiments of Section VII-A. We sweep the full sphere of push directions, 2000 uniformly drawn directions at each of six configurations spread over the workspace, and for each direction record the relative error of both inversions together with the alignment $|\cos \theta|$ between the push and the least observable eigenvector of $\mathbf{P}_{ff}$ at that pose.

Table 4 reports the result. Three observations follow. The median error of the six-dimensional inversion is between 47.7 and 49.5 % at every pose, and roughly half of all push directions incur an error above 50 %, so the failure is not confined to a special direction or a special posture. The point-contact inversion is exact to 4×10^{-13} % over all 12 000 directions and all six poses. And the poses span $\sigma_{\min}(\mathbf{J}_v)$

from 0.0255 to 0.0802 without the six-dimensional error responding, confirming that its error is not a conditioning effect.

Fig. 3 confirms the mechanism quantitatively. Pooling the 12 000 directions, the median six-dimensional error grows monotonically with alignment, from 9.8 % below $|\cos \theta| = 0.2$ to 94.8 % above 0.95, and the correlation between alignment and error is 0.9993. The geometric explanation is therefore not an interpretation placed on the numbers after the fact: it accounts for essentially all of the variance.

Remark 5 (Scope of the correlation). *The least observable direction is obtained from $\mathbf{P}_{ff}$, that is from the same model that produces the error; so the correlation of 0.9993 should be read for what it is: evidence that a single scalar function of the push direction predicts the error, not an independent confirmation. What it excludes is that the error depends on magnitude, on posture, or on conditioning, all of which are varied across the sweep without affecting it. An independent test would identify the least observable direction empirically on hardware and check that it coincides with the one $\mathbf{P}_{ff}$ predicts.*

V. ESTIMATION UNDER VELOCITY-LEVEL ACTUATION

The residual (7) is written in torque, while the actuator accepts velocity. This section shows that the mismatch is apparent, that velocity-level actuation in fact simplifies the estimation problem, and that it changes both the error budget and the nature of the compliance obtained.

Remark 1 already established the structural point: the residual uses τ_{act} as measured through the effort channel, so the servo gain k_v does not enter the estimator and the inner loop need not be identified. What must be known is the map from reported current to torque, which is a calibration rather than a system identification. The remainder of this section works out the quantitative consequences.

A. WHERE THE INTERACTION SIGNAL APPEARS

Simulating the plant (5) under a 6 N push at $\mathbf{q}_{\text{task}}$, and holding $\mathbf{u} = \mathbf{0}$ so that the servo attempts to keep the arm still, the residual (7) splits between its inertial term $\mathbf{M}\ddot{\mathbf{q}}$ and its actuator term $\mathbf{h} - \tau_{\text{act}}$ as reported in Table 5.

For any realistic loop gain the force appears almost entirely in τ_{act} . The servo rejects the disturbance, the arm barely moves, and the interaction is expressed as the torque the actuator must produce in order not to move. Two consequences follow, and they redirect the engineering effort in a way that is not obvious beforehand.

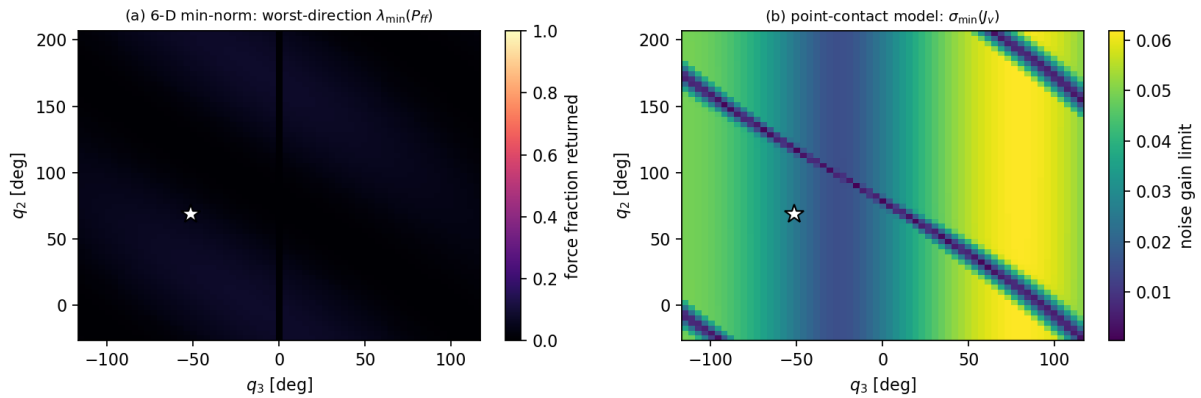
First, the finite-differenced acceleration, normally the delicate quantity in a residual observer, becomes irrelevant: it carries 0.006 % of the signal. Effort spent filtering encoder differences is misplaced on this platform.

Second, the entire error budget transfers to the current-to-torque calibration, where sensitivity is one to one: a 5 % error in the torque constant produces a 5 % error in force. A torque offset is milder, 0.05 N · m per joint producing 3.7 %.

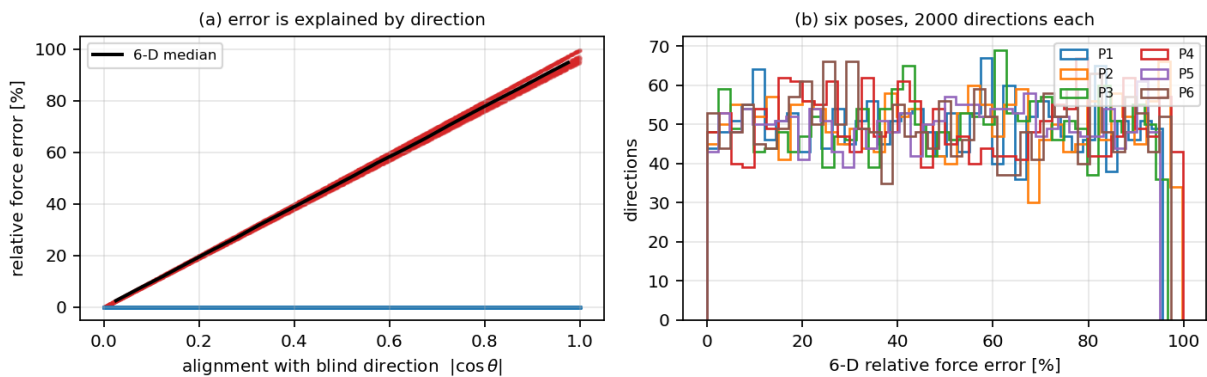
TABLE 3. Workspace sweep over 2197 configurations. The contact force is exactly recoverable everywhere by the point-contact estimator, while the six-dimensional inversion discards half of it on average.

quantity	mean	median	5th percentile	extreme
$\sigma_{\min}(\mathbf{J}_v)$	0.0500	0.0496	0.0127	7.6×10^{-5} (min)
$\text{cond}(\mathbf{J}_v)$	—	6.7	—	29.4 (95th percentile)
$\text{rank } \mathbf{J}_v = 3$	100 % of configurations			
<i>error estimating a unit contact force</i>				
six-dimensional, $\ \hat{\mathbf{f}}_6 - \mathbf{f}\ $	0.5046	0.5110	—	0.9961 (max)
point contact, $\ \hat{\mathbf{f}}_3 - \mathbf{f}\ $	3.6×10^{-15}	7.1×10^{-16}	—	1.4×10^{-12} (max)

Force estimation over the workspace: the 6-D inverse loses force where the point-contact model does not

**FIGURE 2.** Force estimation over a slice of the workspace ($q_1 = 0$, $q_4 = 0.4$ rad). (a) Worst-direction eigenvalue of P_H , the fraction of a pure contact force that the six-dimensional inversion returns as force; close to zero throughout, which is the effect that has been misread as blindness of the mechanism. (b) $\sigma_{\min}(J_v)$, which bounds noise amplification for the point-contact inversion and is the quantity that actually limits it. The star marks the task configuration.

Point-contact error is below 10^{-12} % for every direction and pose

**FIGURE 3.** Directional sweep over 12 000 push directions at six configurations. (a) Relative force error against alignment with the least observable direction: the six-dimensional error (red) is a linear function of alignment with correlation 0.9993, while the point-contact error (blue) lies at zero. (b) Error distributions of the six-dimensional inversion at each pose, which coincide, showing the effect is a property of the architecture rather than of a posture.

B. DETECTION FLOOR

Combining Section V-A with (11) gives the smallest force distinguishable from residual error. A residual error of $0.01 \text{ N} \cdot \text{m}$ corresponds to 0.20 N at the median posture, 0.79 N at the fifth percentile of $\sigma_{\min}(J_v)$, and is unusable at the worst posture. Since the dominant term on a small geared servo is

stiction, of order $0.05 \text{ N} \cdot \text{m}$, the expected floor is one to two newtons at a typical posture. This is adequate for hand guiding with pushes of five to fifteen newtons, and it is the quantity that governs the design.

TABLE 4. Directional sweep. Relative force error over 2000 push directions at each of six configurations.

pose	$\sigma_{\min}(J_v)$	six-dimensional [%]			point contact
		median	p90	max	max [%]
P1	0.0255	47.7	85.7	95.6	4.1×10^{-13}
P2	0.0357	49.5	89.7	99.6	1.9×10^{-13}
P3	0.0597	49.1	86.3	96.5	1.9×10^{-13}
P4	0.0403	48.1	89.6	99.8	2.2×10^{-13}
P5	0.0802	48.8	85.5	94.9	1.5×10^{-13}
P6	0.0303	48.0	87.7	97.3	2.0×10^{-13}

TABLE 5. Distribution of the residual as a function of velocity-loop stiffness, for a 6 N push.

k_v	$\ M\ddot{q}\ $	$\ h - \tau_{\text{act}}\ $	in τ_{act}	displacement
1	0.0175	1.337	98.7 %	200 mm
4	0.0018	1.370	99.9 %	43 mm
20	0.0001	1.301	100.0 %	8 mm
100	≈ 0	1.289	100.0 %	1.6 mm

C. TWO COMPLIANCE REGIMES

Because the servo behaves as a velocity source, the arm does not yield under a push of its own accord, and compliance must be produced by displacing the reference: sense, move the reference, track. This synthetic compliance is limited by loop latency rather than by any physical property.

That description is only half correct, and the missing half is favorable. The actuator torque is bounded, so the force the arm can exert is bounded as well. Sweeping 729 configurations and evaluating the worst direction at each, subject to $|\tau_i| \leq 1.4 \text{ N} \cdot \text{m}$ and supporting the arm's own weight, the maximum exorable force has median 5.88 N, fifth percentile 4.29 N, and minimum 3.93 N. Manual guidance requires five to fifteen newtons. The interaction therefore has two regimes:

- *Below roughly 4 to 6 N.* The servo resists, the arm barely moves, and compliance is synthetic. Loop latency governs how the interaction feels.
- *Above that threshold.* The actuator saturates and the arm is displaced until it reaches a posture where it can hold the load. Compliance is physical and obtained for free.

The transition between the two is not a matter of interpretation. It has a measurable signature: below the threshold the arm follows the reference the admittance commands, while above it the actuator saturates and the arm is carried further than commanded. Table 6 reports the ratio of actual to commanded end-effector displacement in the closed compliant loop under pushes of increasing magnitude.

The transition falls between 5 and 6 N, in agreement with the actuator capability computed independently above, and the two indicators change together: the ratio departs from unity exactly where the saturated fraction goes from none to all.

The arm is thus naturally backdrivable in the regime of interest, which is a safety property rather than a deficiency, and the synthetic loop only has to cover fine guidance and

TABLE 6. The two compliance regimes. Steady-state end-effector displacement, actual versus commanded by the admittance, under a push of increasing magnitude.

$\ f\ $ [N]	commanded [mm]	actual [mm]	ratio	saturated steps
1	8.3	8.6	1.03	0 %
3	24.8	23.7	0.95	0 %
5	41.4	37.8	0.91	0 %
6	49.7	86.0	1.73	100 %
8	66.2	124.5	1.88	98 %
10	82.8	162.3	1.96	98 %
15	124.1	207.8	1.67	99 %

the return to task. Two practical consequences follow. The latency requirement on the control loop is milder than a purely synthetic analysis would suggest, since above the threshold the physics yields without waiting for the loop. And actuator sizing becomes a compliance design variable: choosing a servo whose force capability falls below the expected interaction range confers intrinsic backdrivability with no control effort at all.

Remark 6 (The estimator does not saturate). *One might expect the estimate to saturate together with the actuator. It does not. Under a force ramp with $|\tau_i| \leq 1.4 \text{ N} \cdot \text{m}$ the estimate remains exact across the range, returning 1.000, 5.999, and 13.998 N for applied magnitudes of 1, 6, and 14 N, an error of 0.0 % throughout. The reason is that the residual (7) contains both terms: when the actuator can no longer hold, the arm accelerates and redistributes its posture, and the balance shifts from τ_{act} towards $M\ddot{q}$ and towards a configuration in which the load can be supported. The formulation captures both regimes without modification.*

VI. COMPLIANT CONTROL

A. ADMITTANCE

The estimated contact force drives a first-order admittance that displaces the reference pose from the nominal task pose, with a return spring that restores it when contact ceases,

$$D\nu = \hat{f}_3 - k e_{\text{task}}, \quad (12)$$

where ν is the reference twist, D collects translational and rotational damping, k is the return stiffness, and e_{task} is the displacement of the reference from the task pose. The twist is integrated on the dual pose reference directly, so translation and rotation remain one object.

B. PREDICTIVE CONTROLLER

A nonlinear predictive controller tracks the compliant reference. The state is $x = [q^T \dot{q}^T]^T \in \mathbb{R}^8$, the input is joint torque, and the servo receives the predicted velocity $\dot{q}_1^*$, which matches its native command mode. The optimal control

TABLE 7. Controller formulations under velocity-level actuation. Regulation to a fixed pose from an initial error of 0.16 m, on the plant (5) with torque saturation.

	n_x	needs $\mathbf{M}, \mathbf{h}$	needs k_v	error [mm]	ms/solve
T	8	yes	no	0.180	2.60
V1	8	yes	yes	<i>fails</i>	2.67
V0	4	no	no	0.122	0.086

Error after settling; V0 with command weight tuned. V1 returns NAN_DETECTED at every solve, see text.

problem is

$$\begin{aligned} \min_{\mathbf{x}, \boldsymbol{\tau}} \quad & \sum_{k=0}^{N-1} \|\mathbf{y}_k\|_{\mathbf{W}}^2 + \|\mathbf{y}_N\|_{\mathbf{W}_N}^2 \\ \text{s.t.} \quad & \mathbf{x}_0 = \hat{\mathbf{x}}, \quad \mathbf{x}_{k+1} = \Phi(\mathbf{x}_k, \boldsymbol{\tau}_k), \\ & \mathbf{q}_{\min} \leq \mathbf{q}_k \leq \mathbf{q}_{\max}, \quad \|\dot{\mathbf{q}}_k\|_{\infty} \leq \dot{\mathbf{q}}_{\max}, \\ & \|\boldsymbol{\tau}_k\|_{\infty} \leq \tau_{\max}, \end{aligned} \quad (13)$$

with residual $\mathbf{y} = [\mathbf{e}^\top \dot{\mathbf{q}}^\top \boldsymbol{\tau}^\top]^\top$ built on (6), and Φ the discretized dynamics (1) with $\mathbf{F} = \mathbf{0}$, since the interaction enters through the reference rather than the prediction model. The horizon is $N = 20$ nodes over 1.0 s.

C. WHICH MODEL SHOULD THE CONTROLLER USE?

Formulation (13) takes torque as the decision variable while the actuator accepts velocity, so the predicted velocity is passed to the servo. This is the customary arrangement, but on velocity-actuated hardware it is not the only one, and the alternatives differ in what they require the designer to know. We implemented and compared three.

- **T**, torque level: $\mathbf{x} = [\mathbf{q}, \dot{\mathbf{q}}]$, $\mathbf{u} = \boldsymbol{\tau}$, the dynamics (1), with $\dot{\mathbf{q}}_1^*$ sent to the servo. Requires $\mathbf{M}$ and $\mathbf{h}$. Does not require k_v .
- **V1**, dynamic velocity level: $\mathbf{x} = [\mathbf{q}, \dot{\mathbf{q}}]$, $\mathbf{u}$ the velocity command, the plant (5). Requires $\mathbf{M}$, $\mathbf{h}$ and k_v , that is, identification of the servo loop.
- **V0**, kinematic: $\mathbf{x} = [\mathbf{q}]$, $\mathbf{u}$ the velocity command, $\dot{\mathbf{q}} = \mathbf{u}$, the limit of (5) as $k_v \rightarrow \infty$. Requires nothing: no inertia, no bias term, no servo gain.

Table 7 reports the comparison, and the outcome is not the expected one.

V1 is ill-posed. Every solve fails. The cause is instructive rather than numerical: with a proportional loop model, sustaining a velocity requires a persistent velocity error, so imposing the true actuator bound $|k_v(u_i - \dot{q}_i)| \leq \tau_{\max}$ restricts the command to $|u_i| \leq \tau_{\max}/k_v = 0.07$ rad/s, which is 85 times below the 5.969 rad/s the servo actually reaches. Omitting the bound is worse, since the model then admits torques of 239 N·m, two orders of magnitude beyond the actuator, and the integrator diverges. The defect is the model structure: the real servo loop has integral action, so no persistent torque is needed to hold a velocity. Formulating V1 correctly therefore requires identifying not only a gain but the loop architecture including anti-windup, which is precisely the burden the other two formulations avoid.

V0 needs no model at all and is not worse. Once the command weight is tuned it settles to 0.122 mm against 0.180 mm for T, at 0.086 ms per solve, a factor of thirty faster, with no failures. Its state is half the size and it uses neither the inertia matrix nor the bias term.

This yields a clean separation of concerns, which we regard as the practical recommendation of this paper:

On a velocity-actuated arm the dynamic model is required by the estimator and by nothing else. The controller can be model-free.

The estimator needs $\mathbf{M}$ and $\mathbf{h}$ to form the residual (7), and by Remark 1 it needs no knowledge of the servo loop. The controller needs neither. The only quantities to be identified on hardware are therefore the kinematics, the mass properties, and the current-to-torque constant, all of which serve estimation, and none of which serve control.

Remark 7 (What V0 gives up). *The kinematic formulation cannot express torque bounds, so it cannot by itself prevent commands the actuator will not follow. Two considerations make this acceptable here. The velocity bound is retained and is the binding limit in normal operation, and by Section V-C exceeding the torque bound does not produce a fault but the physically compliant regime, in which the arm yields. If a hard torque bound is required, for instance to protect a payload, the static requirement $\|\mathbf{h}(\mathbf{q}) + \mathbf{J}_v^\top \hat{\mathbf{f}}\|_{\infty} \leq \tau_{\max}$ can be imposed on V0 as a state constraint without reintroducing k_v .*

Table 7 compares the three under regulation. The claim that the controller can be model-free is stronger than a regulation test, however, so Section VII-C repeats the complete compliant interaction loop with V0 in place of T.

Problem (13) is transcribed with CasADi [12] and solved by the real-time iteration scheme [10] in acados [11], with HPIPM [13], a Gauss-Newton Hessian, and a fourth-order explicit Runge-Kutta integrator.

Because $n = 4$, a general SE(3) reference is unreachable and the optimizer settles on a compromise between orientation and translation error. Which compromise it selects is decided by the metric, which is what $\mathbf{J}_\ell^{-1}$ in (6) supplies; Section VII-I isolates its contribution.

D. REAL-TIME IMPLEMENTATION

Reaching real-time rates required one change whose effect is disproportionate to its visibility, and which we report because the failure mode is easy to reach and hard to diagnose.

In an initial implementation the symbolic model was assembled with the graph-valued expression type of CasADi and the forward dynamics written as $\ddot{\mathbf{q}} = \text{solve}(\mathbf{M}, \boldsymbol{\tau} - \mathbf{h})$. On graph-valued operands this does not produce a closed-form expression: it emits a node backed by a numerical QR factorization, whose derivatives are obtained by implicit differentiation of the linear system and which cannot be expanded to scalar form. With a four-stage explicit integrator propagating sensitivities across twenty nodes, that routine is

TABLE 8. Solve time over 1949 consecutive solves. $N = 20$, $n_x = 8$, $n_u = 4$.

build	mean [ms]	max [ms]	rate [Hz]
graph-valued, closed loop	95.4	104.9	10
scalar, run 1	2.53	3.27	395 / 306
scalar, run 2	2.97	5.19	337 / 193

Rate given as mean / worst case.

TABLE 9. Closed-loop estimation from an identical residual under two contact models.

push	estimator	RMSE [N]	corr.	error
in plane $\ f\ = 6.325$ N	six-dim.	0.024	1.0000	0.1 %
	point	0.008	1.0000	0.0 %
normal $\ f\ = 6.000$ N	six-dim.	4.832	0.8069	83.4 %
	point	0.007	1.0000	0.0 %

evaluated approximately eighty times per solve. The measured cost was 101.6 ms, of which 101.5 ms was linearization and 0.13 ms the quadratic program: the optimizer was never the bottleneck.

Rebuilding the identical model on the scalar expression type turns the linear solve into symbolic Gaussian elimination and yields flat generated code, with the timings of Table 8. The controller is unchanged: over a 250-step closed-loop trajectory the two implementations agree to 2.2×10^{-15} N·m in commanded torque and 3.1×10^{-15} in state.

VII. RESULTS

Estimation is reported first against an analytic plant, where ground truth is exact, and then against MuJoCo, which provides an independent implementation of the dynamics including effects absent from (1).

A. FORCE ESTIMATION IN CLOSED LOOP

A hand-guiding scenario is simulated at 100 Hz. The arm holds a task pose, an external force follows a trapezoidal profile, and the compliant loop runs throughout. Both estimators are computed from the *same* residual at every step, so the only difference is the assumed contact model. Two push directions are used: one in the plane of the arm and one normal to it, the latter being the direction identified in Section IV-E.

Table 9 and Fig. 4 report the outcome. For the in-plane push both estimators are accurate, which is the regime in which the conventional inversion has historically been validated. For the normal push the six-dimensional inversion is wrong by 83.4 % while the point-contact inversion remains exact to the discretization floor of the benchmark. In both cases the compliant loop yields under the push and returns on release, with the solver reporting success at all 350 steps.

Table 10 states the same result without reference to any angle, for three directions that name themselves, at four configurations. A push along the arm or along the base axis lies in the plane of the arm and both inversions recover it. A push normal to that plane is recovered in full by the point-contact

TABLE 10. Recovered force for three physically named push directions, as a percentage of the applied magnitude. The point-contact inversion returns 100.0 % in every entry of the table.

pose	six-dimensional inversion [%]			spurious moment
	along the arm	vertical	out of plane	out of plane
P1	99.9	100.0	6.8	0.20
P2	98.5	100.0	17.3	0.06
P3	99.8	100.0	6.7	0.18
P4	97.3	100.0	23.2	0.04

along the arm: radial direction of the end-effector; *vertical*: base axis; both lie in the plane of the arm. *out of plane*: normal to that plane. Moment in N · m per newton applied.

inversion and largely converted into a moment by the six-dimensional one, at every configuration tested.

B. VALIDATION AGAINST AN INDEPENDENT ENGINE

Replacing the analytic plant with MuJoCo and applying a known external force gives a check that does not share the implementation of the dynamics. At the push peak the applied force is [0.00, 4.00, −1.50] N; the six-dimensional inversion returns [0.59, 0.23, −0.83] N and the point-contact inversion [0.21, 3.92, −1.31] N. Over the push window the RMSE against the applied force falls from 2.227 N to 0.157 N, a factor of fourteen.

Remark 8 (Interpretation of the residual error). *The remaining 0.157 N is not estimator error, since the same estimator is exact to 10^{-15} on the analytic plant. It is the model-to-engine gap: joint damping and friction, the discrete constraint solver, and effects not represented in (1). It is the best available proxy for hardware degradation and should be read as a lower bound.*

Remark 9 (Correlation is not accuracy). *In the same experiment the six-dimensional estimate correlates with ground truth at 0.997 while carrying the 2.227 N error. Correlation measures shape and is insensitive to gain, so an agreement metric of this kind can appear healthy while the quantity of interest is substantially wrong.*

C. THE COMPLIANT LOOP WITHOUT A DYNAMIC MODEL

Section VI-C argued from a regulation test that the controller need not carry a dynamic model. We repeat here the complete interaction loop of Section VII-A, identical in every respect except that the kinematic controller V0 replaces the torque-level controller T. The estimator is unchanged and retains the dynamic model, which it requires.

Table 11 reports the outcome. Force estimation is unaffected, as it must be, since the estimator did not change. The compliant behavior is equivalent: the arm yields comparably under the push and returns closer to the task pose on release, in both push directions. The computational cost falls by a factor of twenty-five. Carrying the inertia matrix and the bias term in the controller therefore buys nothing on this platform, while carrying them in the estimator is unavoidable.

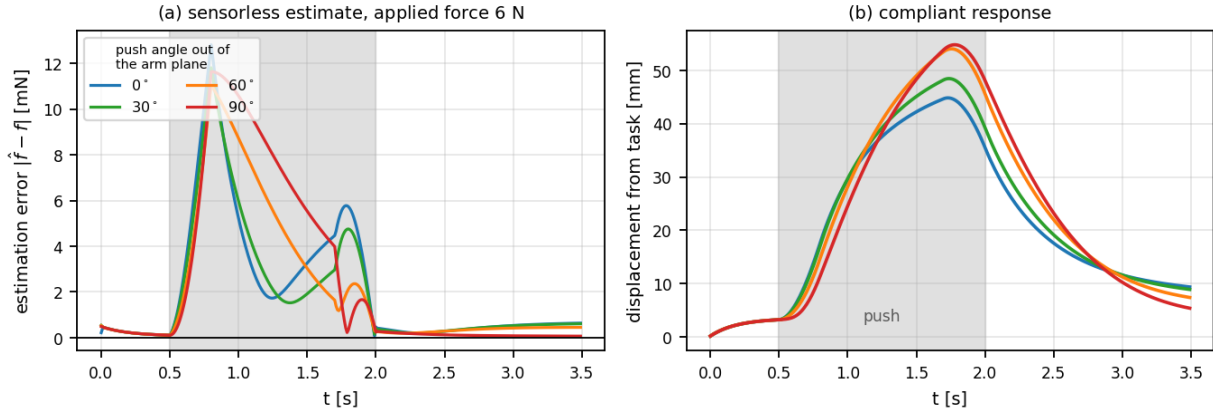


FIGURE 4. Closed-loop interaction under a 6 N push applied at four angles out of the plane of the arm. (a) Error of the point-contact estimate. It stays below 13 mN, that is under 0.25 % of the applied force, at every angle including the one the six-dimensional inversion cannot recover; the residual excursions coincide with the ramps of the force profile and are the discretization floor of the benchmark rather than a directional effect. (b) The resulting compliant behavior: the arm yields by 45 to 55 mm under the push and returns towards the task pose on release, at every angle. The directional dependence of the two inversions is reported in Table 10.

TABLE 11. Complete compliant loop with a model-based controller (T) and a model-free controller (V0). The estimator is identical in both.

push	ctrl	force RMSE [N]	yield [mm]	return [mm]	ms/solve
in plane	T	0.0299	127.1	11.23	2.98
	V0	0.0332	130.7	9.24	0.119
normal	T	0.0042	23.0	3.71	3.05
	V0	0.0045	28.7	1.92	0.123

TABLE 12. Monte Carlo under servo noise, $N = 2000$. Relative force error, as a percentage of the applied magnitude.

noise	six-dimensional			point contact		
	median	p90	p99	median	p90	p99
none	47.2	86.8	97.9	0.0	0.0	0.0
nominal	69.6	149.1	1623	18.0	58.9	353
$\times 2$	89.3	281.2	3194	35.9	118.3	695
$\times 4$	137.3	548.4	6239	71.9	232.8	1402

D. ROBUSTNESS TO SERVO NOISE

Every result reported so far is noise-free, and on a clean bench an exact estimator looks perfect by construction. We therefore evaluate both inversions under the noise the MX-28R actually delivers: current quantization at 3.36 mA per count, Coulomb stiction of $0.05 \text{ N} \cdot \text{m}$ per joint with random sign, which Section VII-F identifies as the dominant term, and twelve-bit encoder quantization. Configurations are drawn uniformly over the joint ranges and contact forces uniformly between 2 and 10 N, over 2000 trials per condition.

Table 12 reports the outcome, and it is the honest expectation for hardware. The point-contact inversion is no longer exact: its median relative error is 18.0 % under nominal noise, against 69.6 % for the six-dimensional one. The advantage is preserved at every percentile and under degraded noise, but both distributions have heavy tails, which the conditioning analysis of Section IV predicts and the next experiment con-

TABLE 13. Effect of rejecting ill-conditioned postures, nominal noise.

$\sigma_{\min}(\mathbf{J}_v)$ threshold	six-dimensional		point contact	
	median	p90	median	p90
0.00	69.6	142.4	18.0	61.3
0.02	70.8	156.2	16.2	42.5
0.04	69.0	146.0	14.5	36.2
0.06	77.7	227.8	12.4	29.5

firms.

Table 13 discriminates between the two error mechanisms. Rejecting postures below a threshold on $\sigma_{\min}(\mathbf{J}_v)$ halves the ninetieth percentile of the point-contact estimator, from 61 % to 30 %, exactly as (11) predicts for an error that is amplified measurement noise. It does nothing for the six-dimensional inversion, whose error is not noise but structural misattribution and therefore cannot be gated away. The two mechanisms identified analytically in Section IV are thus separable experimentally, and monitoring $\sigma_{\min}(\mathbf{J}_v)$ online becomes an actionable recommendation with a measured benefit.

E. DIRECT VERIFICATION OF THE CONDITIONING BOUND

Section VII-D showed that rejecting ill-conditioned postures shortens the tail of the point-contact estimator, which is consistent with (11) but does not measure its exponent. We test the bound directly and through the independent plant.

The arm is held at four postures spanning $\sigma_{\min}(\mathbf{J}_v)$ from 0.011 to 0.092, a factor of nine, selected to be free of contact, clear of the floor, and away from the joint limits. Sensor noise is injected on the current reading, which is where the servo carries it, at one and at three quantization steps of the MX-28R. For each posture we estimate the force sample by sample and record the dispersion of the estimate, which is the quantity (11) bounds.

Table 14 reports the outcome. Fitting $\log \text{std}(\hat{\mathbf{f}})$ against $\log \sigma_{\min}(\mathbf{J}_v)$ gives a slope of -0.872 with $R^2 = 0.980$ at the

TABLE 14. Noise amplification through the MuJoCo plant. Dispersion of the point-contact estimate at four postures, with sensor noise injected on the current reading.

$\sigma_{\min}(\mathbf{J}_v)$	$\text{std}(\tau)$	$\text{std}(\hat{\mathbf{f}})$	bound (11)	ratio
<i>current noise $\sigma = 3.4$ mA (one quantization step)</i>				
0.0108	0.0039	0.183	0.361	0.51
0.0204	0.0036	0.096	0.175	0.55
0.0435	0.0037	0.044	0.085	0.51
0.0918	0.0035	0.030	0.038	0.77
<i>current noise $\sigma = 10$ mA (three quantization steps)</i>				
0.0108	0.0102	0.451	0.945	0.48
0.0204	0.0103	0.250	0.504	0.50
0.0435	0.0102	0.128	0.233	0.55
0.0918	0.0102	0.085	0.111	0.76

lower noise level and -0.790 with $R^2 = 0.987$ at the higher one, against the predicted -1 . The dispersion therefore scales essentially inversely with $\sigma_{\min}(\mathbf{J}_v)$ across a ninefold range, at two noise levels, and through a plant whose damping, armature, and dry friction are absent from the estimator model.

Two details are worth stating. The measured dispersion is consistently below the bound, at 0.48 to 0.77 of it, which is as it should be: (11) is an upper bound attained only when the residual error aligns with the least observable direction. And the fitted slope is slightly shallower than -1 for the same reason, since the alignment varies with posture.

Remark 10. *The noise here is modeled rather than measured: MuJoCo is deterministic, so at equilibrium the actuator torque is exactly constant and quantization alone produces zero variance. What this experiment establishes is that the bound governs the estimator when it is driven through the full nonlinear plant, not merely in the algebraic setting of Section IV. Replacing the modeled noise with a real servo's is the remaining step, and it requires only a static measurement with the actuators disabled.*

F. SENSITIVITY TO MODEL AND CALIBRATION ERROR

Section V places the error budget on the current-to-torque calibration, and Section VII-D shows stiction dominating the noise. Both statements can be made precise by perturbing the validated model one parameter at a time and recording the spurious force the estimator then reports under no contact, which is the quantity a deployment would have to distinguish from a real interaction. Table 15 reports it at the worst of six configurations selected by sweeping the workspace for maximum sensitivity.

Two features of Table 15 matter for deployment. Torque offsets dominate: 0.02 N·m already produces 0.67 N, whereas a 20% error in a link mass produces 0.42 N and a 5% error in the torque constant only 0.19 N. The asymmetry is geometric. A gravity error lies in the well-conditioned directions of $\mathbf{J}_v$ and is largely absorbed by the least-squares projection of (10), whereas an offset is not, which is why stiction rather than mass modeling sets the floor. The estimate is also insensitive to the mass of the first link, correctly so, since that link rotates

TABLE 15. Spurious force reported under no contact when a single model or calibration parameter is perturbed, at the most sensitive of six configurations.

perturbation	spurious $\ \hat{\mathbf{f}}_3\ $ [N]
inverted torque sign	7.38
torque constant $\times 2$	3.69
torque offset 0.05 N·m	1.66
stiction 0.05 N·m	0.90
torque constant $+20\%$	0.74
torque offset 0.02 N·m	0.67
link 4 mass $+20\%$	0.42
joint zero shifted 0.10 rad	0.33
torque constant $+5\%$	0.19
link 3 mass $+20\%$	0.24
link 1 mass $+20\%$	0.00

TABLE 16. Compliant versus rigid guidance over $N = 6$ paired conditions. Both arms visit the identical conditions, so the paired difference is the appropriate statistic.

cond.	guidance force [N]			following error [mm]		
	rigid	compl.	Δ	rigid	compl.	Δ
1	3.59	1.96	+1.62	44.8	24.5	+20.3
2	4.13	3.68	+0.45	51.6	45.9	+5.7
3	3.12	1.63	+1.49	39.0	20.2	+18.7
4	3.46	1.40	+2.06	43.2	17.4	+25.8
5	3.98	3.02	+0.96	49.7	37.6	+12.1
6	4.62	4.33	+0.29	57.7	54.0	+3.7
mean	3.82	2.67	+1.15	47.7	33.3	+14.4
s.d.	0.49	1.09	0.64	6.1	13.6	8.0

about a vertical axis and gravity exerts no torque about it.

A practical consequence follows for calibration. Because gravity torques on this arm are of order 0.3 N·m, a no-contact null test cannot resolve the torque constant: a 5% error in it is indistinguishable from the residual bias of a well-modeled arm. The constant must therefore be identified under a known load rather than at rest.

G. COMPLIANCE VERSUS RIGID BEHAVIOR

Six guidance conditions, each a task pose and a guidance direction, were run twice in MuJoCo, once compliant and once rigid (Table 16).

The design is paired, since both arms visit the identical set of conditions, so the between-condition variance is common and cancels in the difference. The paired differences are $+1.15 \pm 0.64$ N in guidance force, giving $t = 4.42$, and $+14.4 \pm 8.0$ mm in following error, giving $t = 4.43$, against a two-sided critical value of 2.571 at five degrees of freedom, favorable in all six conditions. The unpaired summaries, 3.82 ± 0.49 N against 2.67 ± 1.09 N, have overlapping dispersions and would not support the claim at $N = 6$: the effect is real and the unpaired presentation discards the information that establishes it.

H. TRACKING, REGULATION, AND COMPUTATION

Regulating to a fixed pose from an initial error of 0.160 m converges to 1.65 mm with success at all 250 steps. On a slow

TABLE 17. Metric ablation. Steady-state position error under a reference rotated by α , making it unreachable.

α [deg]	coupled [mm]	decoupled [mm]	advantage
0	1.038	1.050	+1.1 %
15	1.763	1.779	+0.9 %
30	2.472	2.492	+0.8 %
45	3.646	3.691	+1.2 %
60	6.120	6.220	+1.6 %
75	10.794	11.020	+2.1 %
90	17.729	18.246	+2.8 %

screw trajectory the controller achieves 6.82 mm position RMSE (maximum 7.40 mm) and 4.79° orientation RMSE (maximum 6.35°). Solve times appear in Table 8; we give two runs of the identical build because a single timing run is not a measurement, and they differ by 17 % in the mean and 59 % in the maximum.

I. ABLATION OF THE POSE METRIC

The coupled error (6) differs from a decoupled one only through $\mathbf{J}_\ell(\varphi)^{-1}$, and since $\mathbf{J}_\ell^{-1} \rightarrow \mathbf{I}$ as $\|\varphi\| \rightarrow 0$ the two coincide in the small-error limit; a comparison conducted there is a tautology rather than an experiment. The relative difference between the two translational errors grows linearly at 0.38 % per degree of orientation error, reaching 1.9 % at 5°, 16.9 % at 45°, and 34.1 % at 90°.

We therefore ablate where the metrics genuinely differ. Both controllers receive identical cost structure, weights, and solver settings and differ only in the residual. The reference orientation is rotated by α , rendering the target unreachable, which is the normal operating condition of a four-joint arm rather than a contrived one; the initial error is large and the transient is included.

Table 17 reports the result. The coupled metric is superior and its advantage grows monotonically with α , with orientation error indistinguishable and no solver failures. At $\alpha = 0$, where the reference is reachable, the two tie as predicted, which serves as a positive control.

The effect is real but small. Where the cost functions differ by 34 % the closed loop differs by 2.8 %, because torque saturation and the reachable manifold dominate the compromise. We therefore do not advance the coupled error as a performance claim. Its merit is that it supplies the geometrically consistent compromise by construction and at no computational cost, both branches requiring 2.48 ms per solve.

VIII. DISCUSSION

A. PRACTICAL IMPLICATIONS

The correction of Section IV is inexpensive to adopt. It replaces one linear solve with another of smaller dimension and requires no additional sensing, calibration, or tuning. For any application in which the contact location is known by construction, and hand guiding at the end-effector is the archetype, the six-dimensional inversion should not be used on an arm with fewer than six joints.

The analysis also reframes the platform. A four-joint arm sensing contact force through its own actuators is not operating at the margin of feasibility: by Proposition 2 it has one measurement more than the problem requires. The quantity a designer should examine when placing a task is $\sigma_{\min}(\mathbf{J}_v)$, mapped in Fig. 2(b), not any index derived from the rank deficit of the full Jacobian. Velocity-level actuation, in turn, removes the need to identify the servo loop and renders the acceleration term negligible, concentrating the design effort on a single calibration.

B. LIMITATIONS

Simulation only. Every number reported here is obtained in simulation. The model is exact against MuJoCo to 10^{-10} and MuJoCo serves as an independent plant, but neither substitutes for the physical arm. Two quantities are expected to degrade on hardware: the actuator-torque proxy from servo current, which carries stiction and gearbox losses absent from (1), and the finite-differenced acceleration, which will be noisier, although Section V shows the latter matters little. The 0.157 N model-to-engine gap is the best available proxy and is a lower bound.

Known contact point. As stated in Remark 4, the point-contact model requires the contact location. Contact along the links, or contact transmitting a moment through a grasped object, lies outside the analysis.

Conditioning near singularities. Although $\text{rank } \mathbf{J}_v = 3$ throughout the sampled workspace, $\sigma_{\min}(\mathbf{J}_v)$ reaches 7.6×10^{-5} , where (11) predicts severe amplification. A deployment should monitor $\sigma_{\min}(\mathbf{J}_v)$ online, which is inexpensive, and either avoid those postures or widen the admittance dead band there.

Friction is not observed by either route. The servo current measures motor torque upstream of the gearbox, so gearbox friction is unmodeled whether the applied torque is read from current or inferred from the velocity loop. This sets the irreducible floor and is the quantity most likely to dominate on hardware.

Statistical scope. The compliance result rests on six paired conditions in a single simulator, and the timing figures vary by 17 % between two runs of an identical build. We report both rather than the more favorable one.

IX. CONCLUSION

On manipulators with fewer actuated joints than task dimensions, the accuracy of sensorless interaction sensing is governed by the contact model the estimator assumes rather than by the rank of the Jacobian. Inverting the momentum residual for a full six-dimensional wrench is under-determined and silently converts contact force into spurious moment, discarding 50.5 % of the applied force on average over the workspace studied and up to 99.6 %. Committing to a point contact at a known location renders the problem over-determined, and the force is recovered exactly whenever $\text{rank } \mathbf{J}_v = 3$, which holds in 100 % of the sampled workspace, leaving numerical conditioning as the binding limitation. Against an indepen-

dent physics engine the correction improves force RMSE by a factor of fourteen.

Velocity-level actuation, commonly regarded as an obstacle to sensorless estimation, is instead a simplification. The servo reports the torque it applies, so the residual is invariant to the inner loop and no identification of it is required; the finite-differenced acceleration carries 0.006 % of the signal and becomes irrelevant; and a kinematic controller matches a model-based one throughout the compliant loop at one twenty-fifth of the cost, which confines the dynamic model to the estimator. Under realistic servo quantization and stiction the corrected estimator attains 18.0 % median relative error against 69.6 %, and gating on $\sigma_{\min}(\mathbf{J}_v)$ halves its tail while leaving the six-dimensional error untouched, separating amplified noise from structural misattribution. Finally, because the actuator saturates below the force range of manual guidance, physical compliance is obtained for free and the synthetic loop need only cover fine guidance and the return to task.

Two lines follow. The first is hardware validation, for which Section VII-F identifies the decisive measurement: since a no-contact null test cannot resolve the torque constant on an arm whose gravity torques are of order $0.3 \text{ N} \cdot \text{m}$, the constant must be identified under a known load, and the conditioning prediction of (11) is then tested by repeating that measurement in a well-conditioned and an ill-conditioned posture. The second is extending the contact model to an unknown contact location along the links, which would restore whole-arm interaction while preserving the dimensional argument that makes the approach work.

ACKNOWLEDGMENT

The authors thank Universidad Tecnológica Indoamérica for its support of this research, and the MIST Research Center for its contributions.

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